

Lesson 8 — Meet the Color Sensor

Matches LEGO Teacher Portal lesson: "Spin and Grin".

Objective (I can...): I can use the color sensor to tell what color it sees. **CS concept:** How a sensor works — the color sensor reads information from the world · **Builds on:** Lesson 7 (we know the motor; now meet the other part).

AI Connection (2 min, read aloud): A self-driving car's camera "sees" colors and signs; your color sensor "sees" colors too — that's how a machine takes in information. *Ask the group:* "Who does a self-driving car help? Could it ever worry someone?"

Materials: LEGO CS & AI kit + device per pair, a set of big color cards (red, green, blue, yellow). **Model reuse:** Hardware machine from Lesson 6; snap the color sensor on as a tiny change if it isn't already — no full rebuild.

60-minute flow - Warm-up (10): "Color hunt" — coach calls a color, kids touch something that color in the room, fast. Then: "How does a machine know a color if it has no eyes?" (It has a sensor.) - **Concept + demo (8):** Attach/point to the color sensor. Hold a color card to it and show the live color picture change on screen. Swap cards; the picture follows. - **Build/Code (25):** Pairs watch the live color reading as they hold up different cards, and code a simple "when I see red, light up red" style reaction so the machine shows what color the sensor sees. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Find a color the sensor gets confused by (a tricky in-between color); talk about why it's hard. - **Share + tidy-up (10):** Pairs show the sensor naming two colors correctly. Model stays built.

Pair roles: Driver codes and runs, Navigator holds color cards to the sensor and reads the on-screen color. Swap at the wiggle break. (*Trio rotates code/hold/read.*)

Differentiation: *Support* — use just two very different colors (red and blue). *Extension / fast-finisher* — make a different light or sound for three colors. *Reteach* — watch the live color picture while slowly moving one card in and out.

Exit check: Pair can show the color sensor correctly telling two different colors apart. **At-home (optional):** Find a machine at home that senses something (a light that turns on by itself); tell a grown-up what it notices. ~5 min.